

Approximating function values - 6.4

Recall : $R_k(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n - T_k(x)$

$$= \sum_{n=k+1}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n$$

We said there's some c between a and x
such that $R_k(x) = \frac{f^{(k+1)}(c)}{(k+1)!} (x-a)^{k+1}$

If $|f^{(k+1)}(c)| \leq M$, then $|R_k(x)| \leq \frac{M}{(k+1)!} |x-a|^{k+1}$.

Example: Approximate $\sin(1)$ using a fifth degree Taylor polynomial. How large can the error be?

$$\rightarrow \sin(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$$

$$\text{so } T_5(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!}, \quad \text{thus the}$$

$$\text{approximation is } \sin(1) \approx T_5(1) = 1 - \frac{1}{3!} + \frac{1}{5!}$$

$$= 1 - \frac{1}{6} + \frac{1}{120} = \frac{120}{120} - \frac{20}{120} + \frac{1}{120} = \frac{101}{120}$$

$$\sin(1) \approx \frac{101}{120}.$$

$$\text{And } |R_5(1)| \leq \frac{1}{(5+1)!} |1|^{5+1} = \frac{1}{6!} = \frac{1}{720} \approx .0014$$

(In fact, error is at most $|R_6(1)| \leq \frac{1}{7!}$ because $T_5 = T_6$.)